TC3000J
Technical Communication

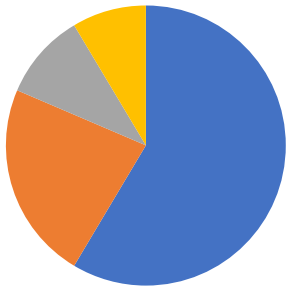
Data Visualisation and Reporting on Visualised Data

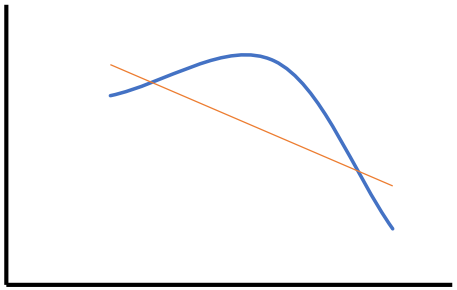
Dr. Xi Liu

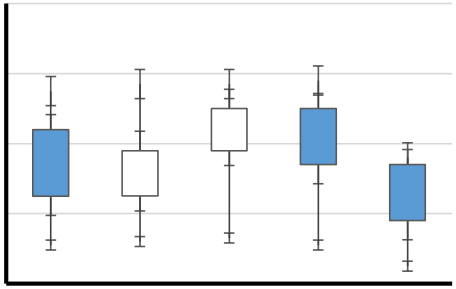
Adjunct Lecturer

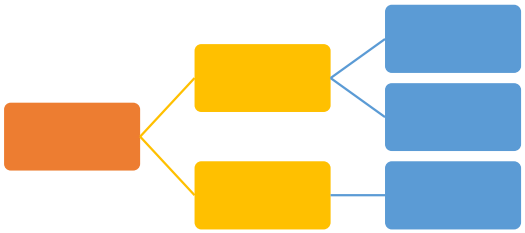
Global College, Shanghai Jiao Tong University, Shanghai, China

Data Visualisation









What is Data Visualization?

Data and information visualization "[…] is the practice of **designing** and **creating easy-to-communicate** and **easy-to-understand** graphic or visual **representations** of a large amount of […] **data** […]."(Cole, 2015)

https://en.wikipedia.org/wiki/Data_and_information_visualization#Overview
Cole, Nussbaumer, Knaflic, "Storytelling with Data: A Data Visualization Guide for Business Professionals" (2015).

Case 1

Failure analysis of fuel cell operation data shows that the top causes of failure are:

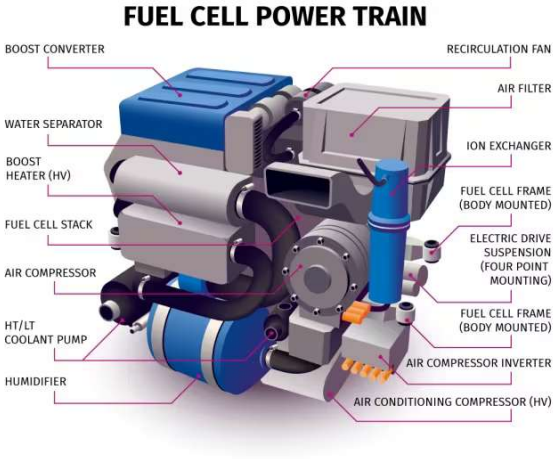
- Membrane Flooding 33%
- Durability 12%
- Membrane Drying 21%
- Leakage 20%
- CO Poisoning 15%
- Others 11%

Case 2

Second year EEE students (**84**) had an exam in *control strategies*, and their results are listed in the table:

Name	Score
A	47
B	95
C	90
D	57
E	93
F	82
G	73
H	99
I	84
J	56
K	77
L	63
M	51
N	61
…	…

Case 3



https://www.dsm.com/engineering-materials/en_US/insights/blog/material-solutions-increase-tank-to-wheel-efficiency-reliability-of-fuel-cells.html
https://ntrs.nasa.gov/api/citations/20170009216/downloads/20170009216.pdf

Why Does It Matter?

Importance

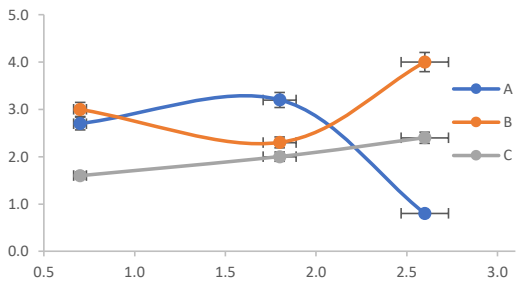
Data is meaningless without thorough interpretation and correct presentation.

Visual aids make the data intuitive and efficient: *a good plot is worth a thousand words.*

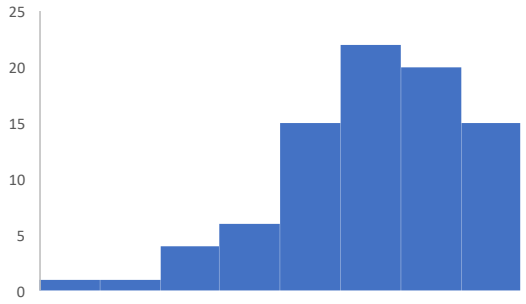
It is valuable to draw the right conclusions from data and have them communicated accurately.

Data visualisation is a must-have skill needed at every stage of data handling.

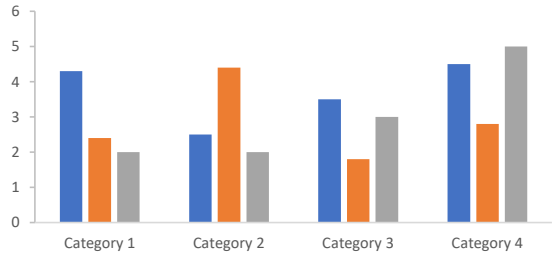
Common Methods and Applications



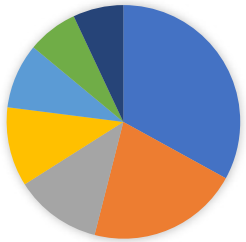
Scatterplot:
Relationship between two numerical variables.



Histogram:
Distribution of numerical data.



Column chart:
Compare categories with a specific order.



Pie chart / Treemap:
General sense of the part-to-whole relationship.

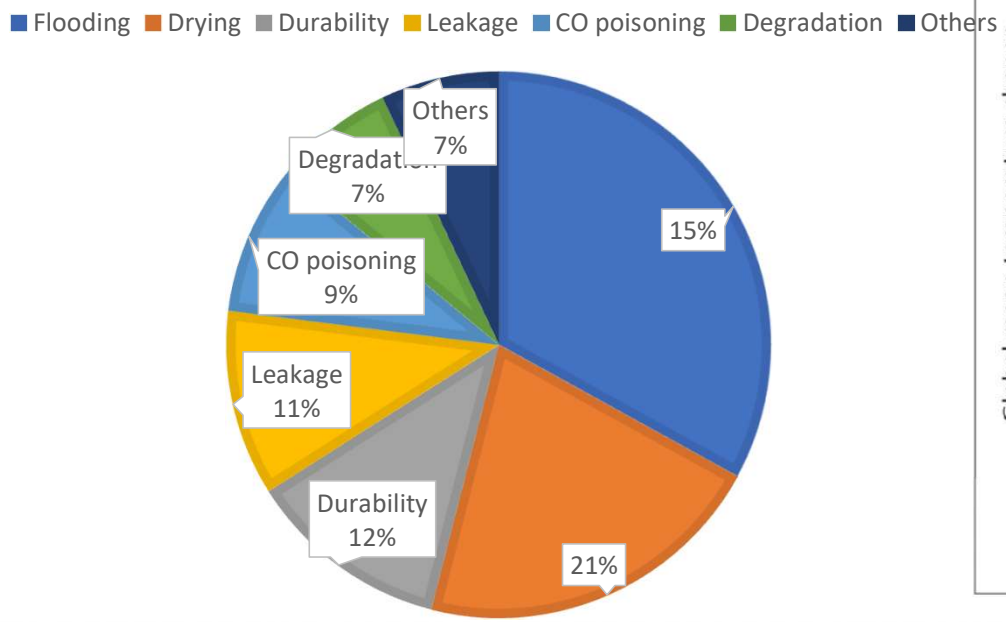
Process Diagram/Flowchart:
Graphical representations of steps
How a process is regulated

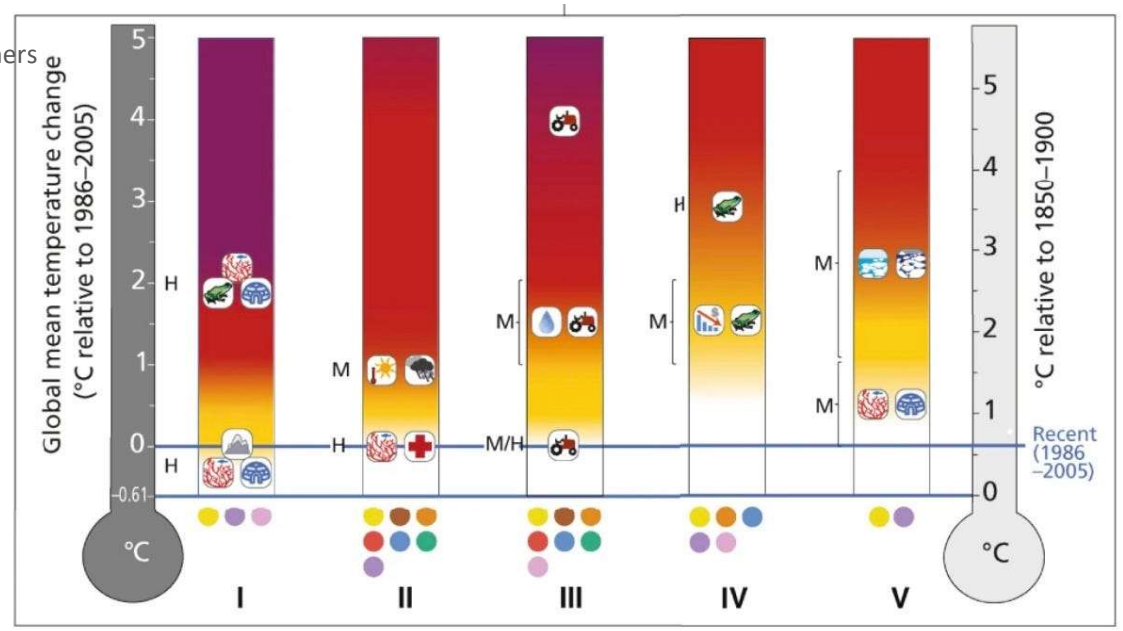
Component	T	P	V
A	70	200	299
B	80	300	800
C	75	249	210
D	90	400	208
E	65	1000	490

Table:
Summarise specific details.
Quantitative information.
BOM, FMEA etc.

Cautions

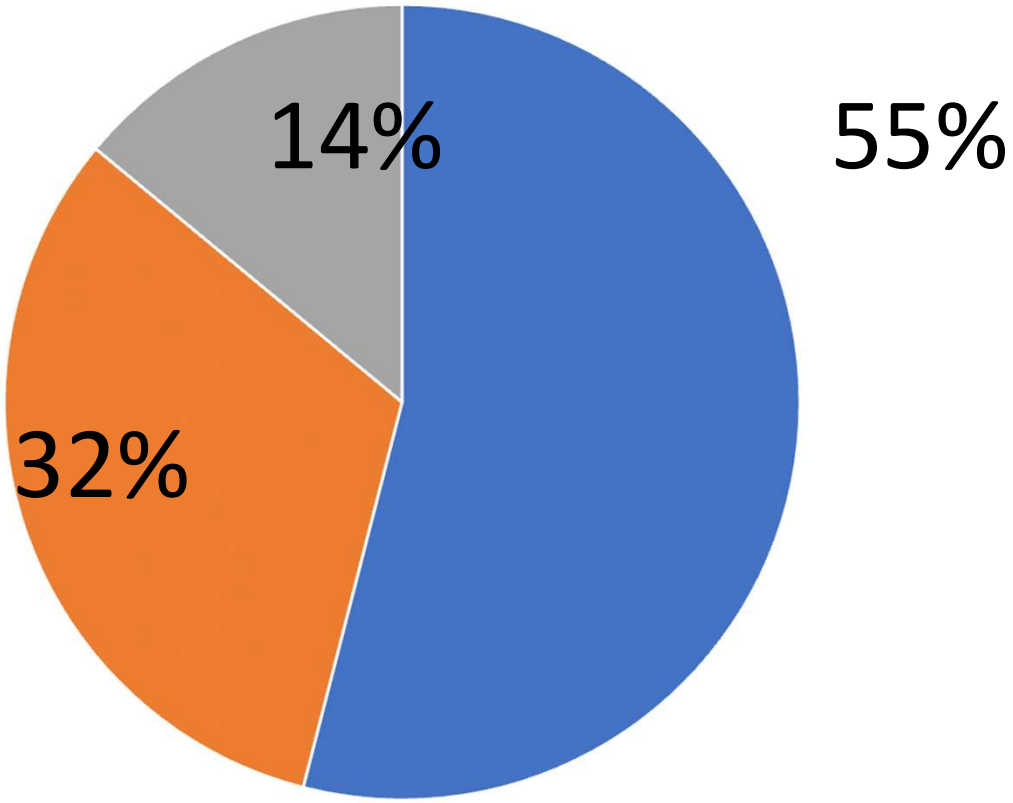
To avoid:



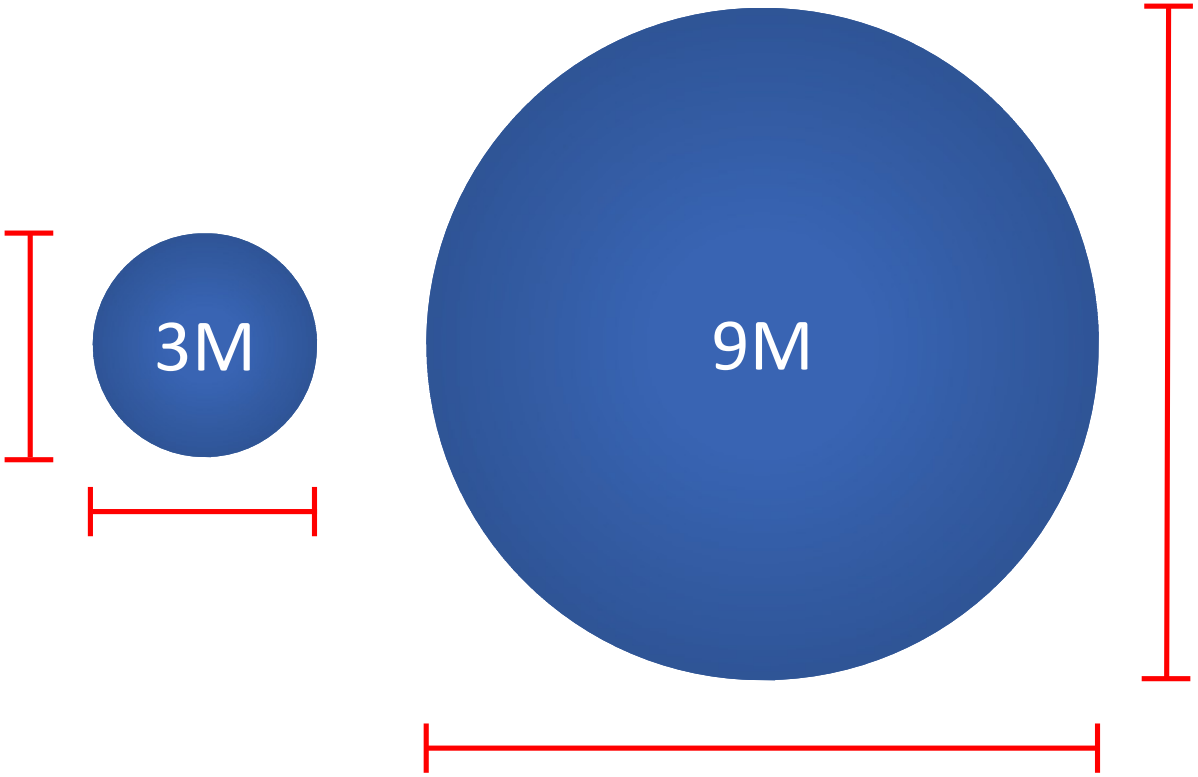


https://filwd.substack.com/p/what-makes-a-data-visualization-confusing

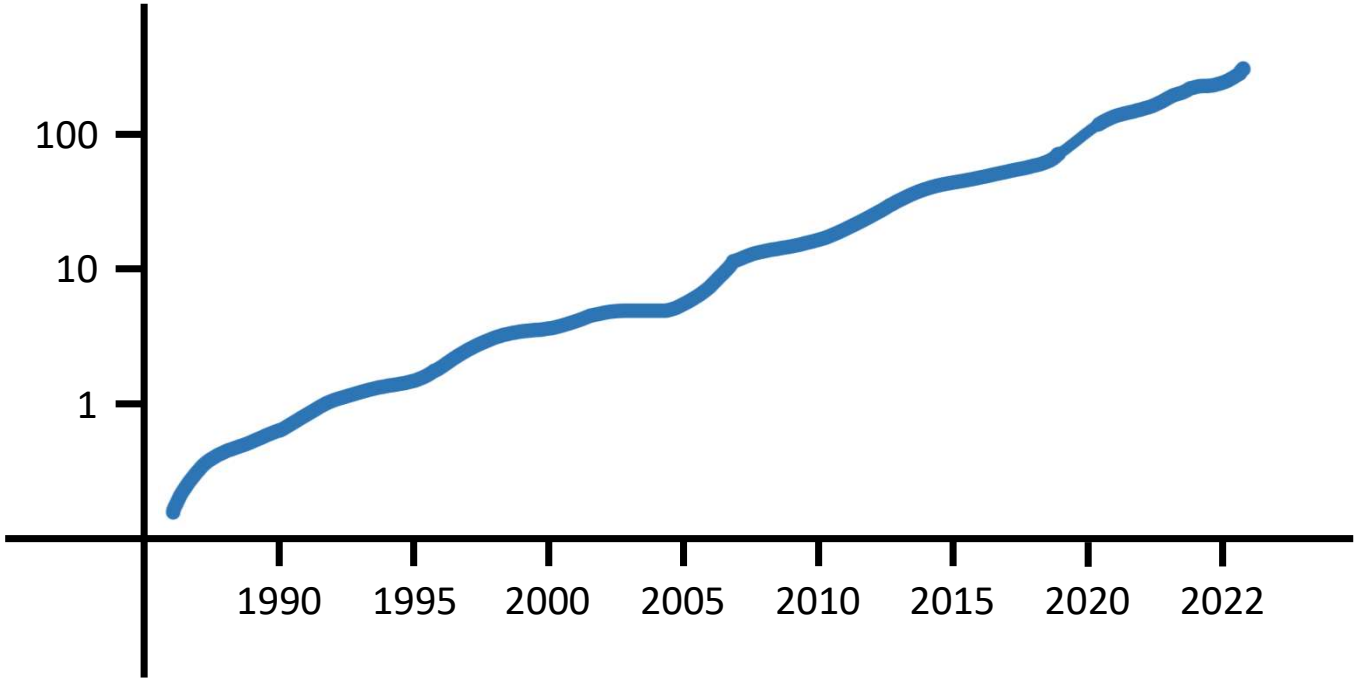
What is Wrong With This Chart?



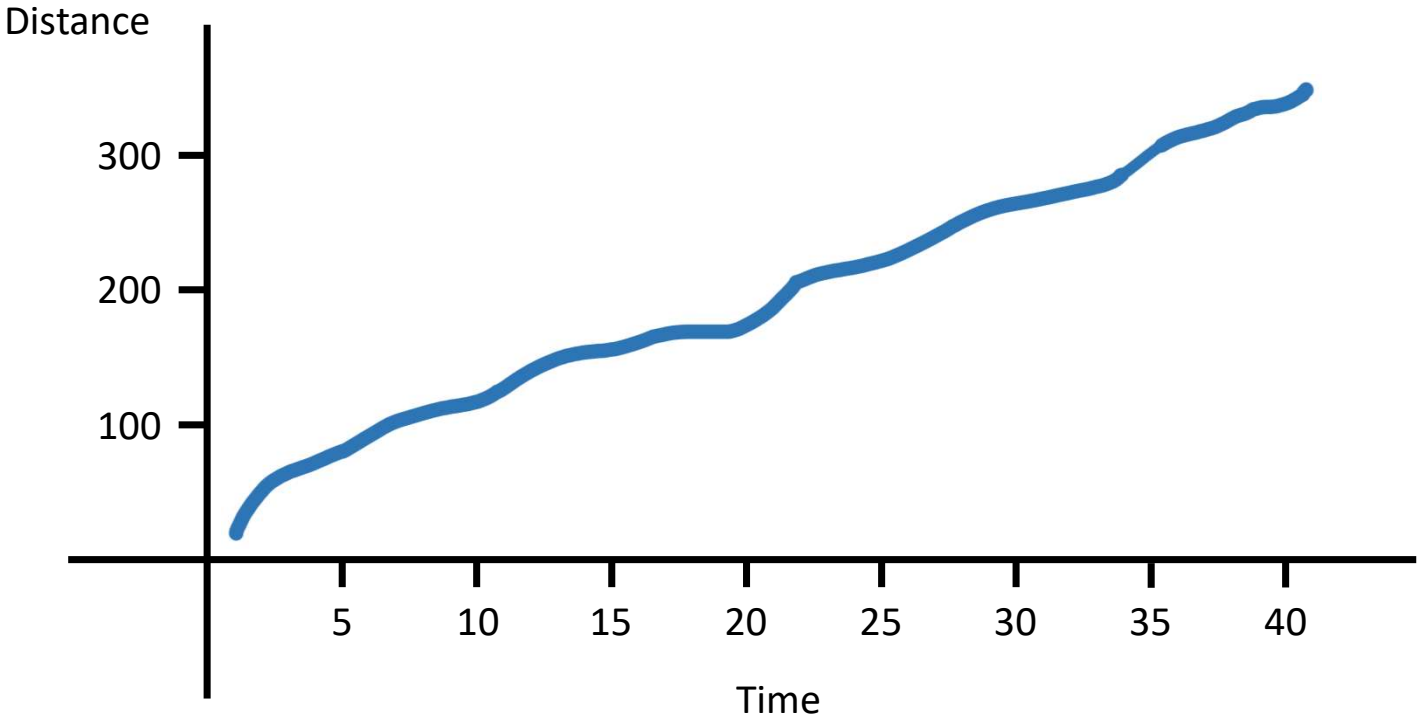
What is Wrong With This Diagram?



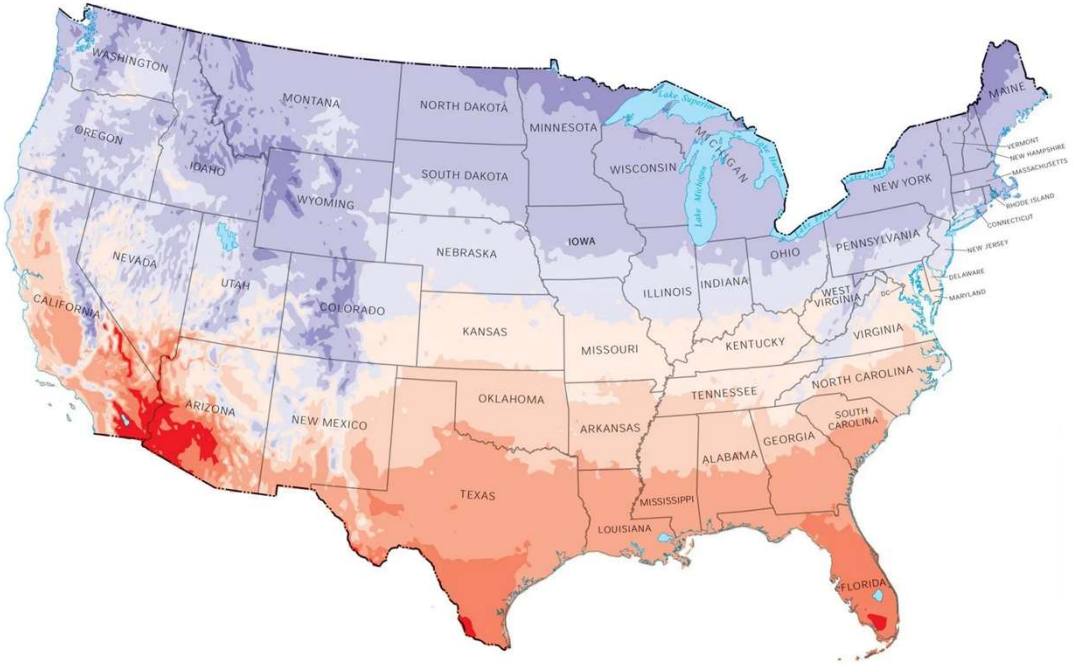
What is Wrong With This Graph?



What is Wrong With This Graph?



What is Wrong With This Diagram?



https://gisgeography.com/us-temperature-map/

What Else is Wrong With This Diagram?

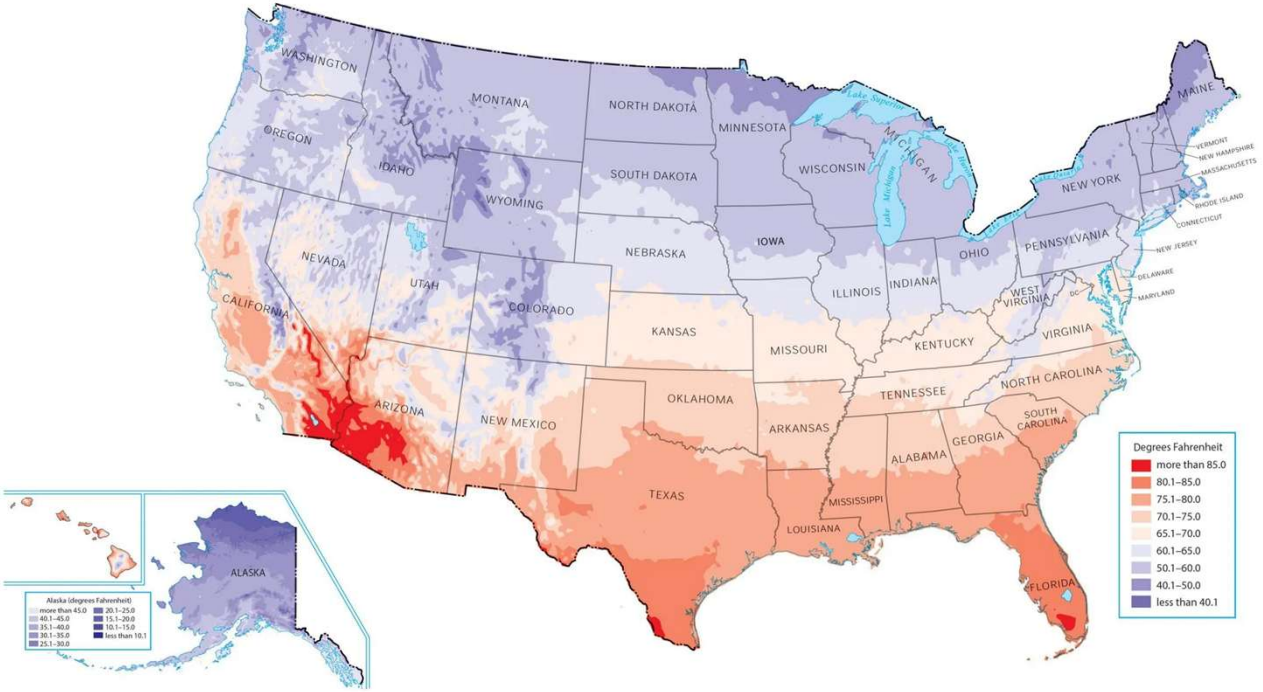


Figure 2. Mean annual daily maximum temperatures in the United States

- Layout
- Accuracy
- Proportions
- Scale and intervals
- Variables
- Units of measurement
- Elements
- Completeness
- Precision

https://gisgeography.com/us-temperature-map/

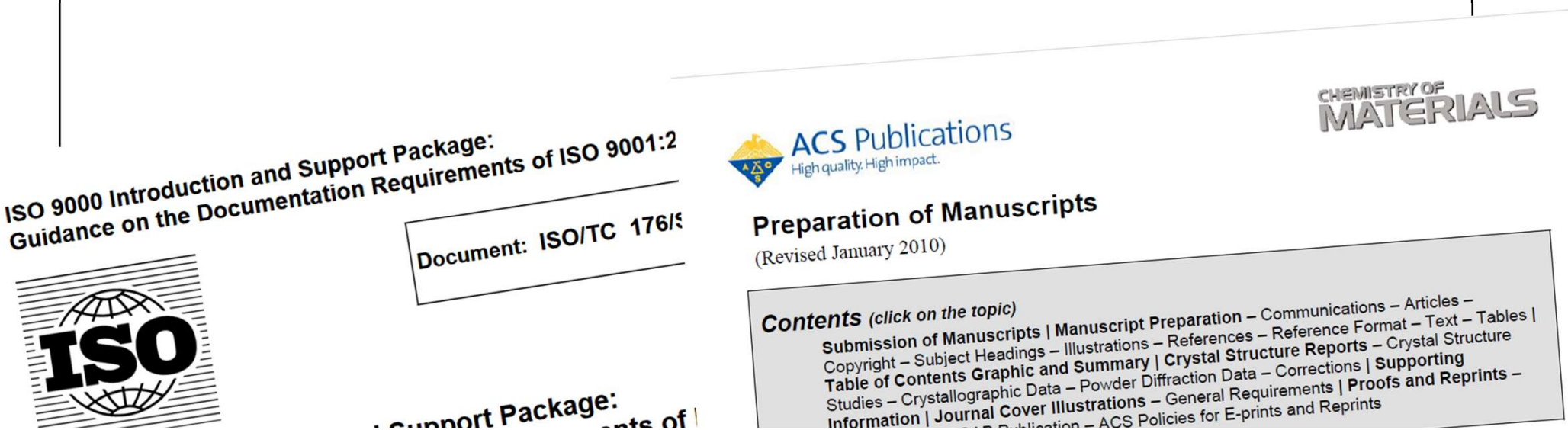
Design Principles

Content
• Use the right graphs/charts for the particular purpose. • Contains “manageable” amount of data • Representing data: Accuracy, Proportions, Scales and Intervals, Variables, Units of Measurements, Completeness • Caption • Key/legend • Error bars. • Use the right tools. • Intended to be integrated with text.

Design Principles

Format

- Colours.
- Labels and units of measurement.
- Spacing and positioning.
- Make sure the fonts, units, ranges are of convenient dimensions and are consistent.
- Check out the requirements.



Reporting on Visualised Data

Referring to a Chart

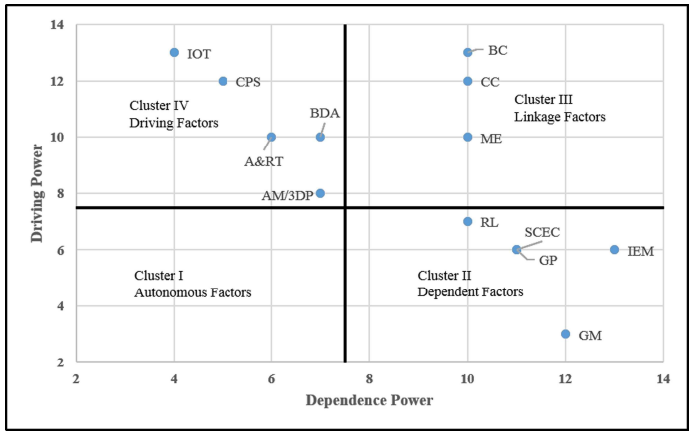


Fig. 4. Results of MICMAC analysis.

A cross-impact matrix multiplication analysis (MICMAC) is mainly used for evaluating the driving and dependence power of factors (Shukla and Shankar, 2022). Three groups of factors are found after the implementation of the MICMAC method and insights about these factors are described considering their driving and dependence power as shown in Fig. 4. Autonomous factors are not present in the analysis, which denotes that all the considered Industry 4.0 and GSC practices are significant. Industry 4.0 technologies like IoT, CPS and A&RT come under the category of independent factors on which the performance of GSCM depends, as they are root causes behind the dependent factors. Among the various driving/independent factors, IoT is a key factor, which has the highest driving power of 13 and dependence power of 4 (Table 6). Therefore, it is placed in the cluster IV with a given dependence and driving power. Blockchain (BC), cloud computing (CC) and ME are found as linkage variables, which can affect other factors in the system as well as receiving the reaction effect. These factors consist of strong driving and dependence power. Blockchain has the highest driving power of 13 and dependence power of 10 and is positioned in cluster III. ME is interlinked with most independent and dependent factors, since it consists of smaller mechanical, optical and electronic devices that are manufactured through Industry 4.0 technologies (Feldmann et al., 2010).

Referring to a Table

The final test conditions are shown in Table 1. Here, seismic waves include Chichi 1999-TCU045, Imperial valley 1979-USGS station 5115, Kobe 1995-Kakogawa CUE90, Northridge 1994–090 CDMG station 24,278 and Landers 1992–000 SCE station 24, all of which come from PEER strong motion database. During the test, the amplitude is increased step by step, and it is stopped when abnormal conditions occur, such as the violent vibration of reaction frame. The sampling interval, as recommended by Reuter [52], is set at 1 ms. Ultimately, 320 signals, each signal including displacement, electrical command, force, etc., were obtained, of which white noises, sine waves, chirp signals, and seismic waves each have 111, 120, 14, and 75. And for the convenience of comparison, each signal was truncated to 32 s.

Table 1. Information on test conditions.

Ind.	Type	Load Mass (kg)	Bandwidth (Hz)	Amplitude	Number
1	white noise	585, 850	1, 2, 4, 6, 8, 10	$0.25d_{max}$, $0.5d_{max}$, $0.75d_{max}$, d_{max},	111
2	sine wave	1050, 1250 1450, 1650			120
3	chirp signal		10		14
4	seismic wave		—	0.01 g, 0.02 g, 0.05 g	75

Referring to Equations

Assuming that the solution of Eq. (1) has the form:

$$q(x,t) = y(z)e^{i\phi(x,t)}, \tag{2}$$

$$z = x - ct, \tag{3}$$

$$\phi(x,t) = -\kappa x + \omega t + \theta, \tag{4}$$

where $y(z)$ denotes shape of the pulse, $c, \phi(x,t), \kappa, \omega$, and θ represents the speed of the soliton, the phase component, the soliton frequency, the soliton wave number and the phase constant respectively. Substituting Eq. (2) into Eq. (1) and after separating real and imaginary parts, the following expressions are obtained

$$(a + 3\delta\kappa)y'' - (\omega + a\kappa^2 + \delta\kappa^3)y + (b - \lambda\kappa)y^3 = 0, \tag{5}$$

$$\delta y''' - (c + 2a\kappa + 3\kappa^2\delta)y' - 3\lambda y^2 y' = 0. \tag{6}$$

Let $a = -3\delta\kappa, \omega = 2\delta\kappa^3, b = \lambda\kappa$ in Eq. (5). Then Eq. (6) can be written as:

$$\delta y''' - (c - 3\delta\kappa^2)y' - 3\lambda y^2 y' = 0. \tag{7}$$

Integrate Eq. (7) and taking constant of integrationto be zero, yields:

$$\delta y'' - (c - 3\delta\kappa^2)y - \lambda y^3 = 0. \tag{8}$$

Writing Guidelines

- Signaling words
- Appropriate data type
- Quantities
- Figure number
- Variables
- Units of measurement